

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Logistic Regression

Goals of this week

- Logistic Regression
- Gradient Descent for Logistic Regression
- Measures of performance and cross-validation

Goals

- What is Classification?
- Why not use Linear Regression?
- Logistic Regression
- Odds
- Maximum Likelihood Estimation
- Logistic Regression optimization
- Gradient Descent and Logistic Regression
- Misc. Logistic Regression

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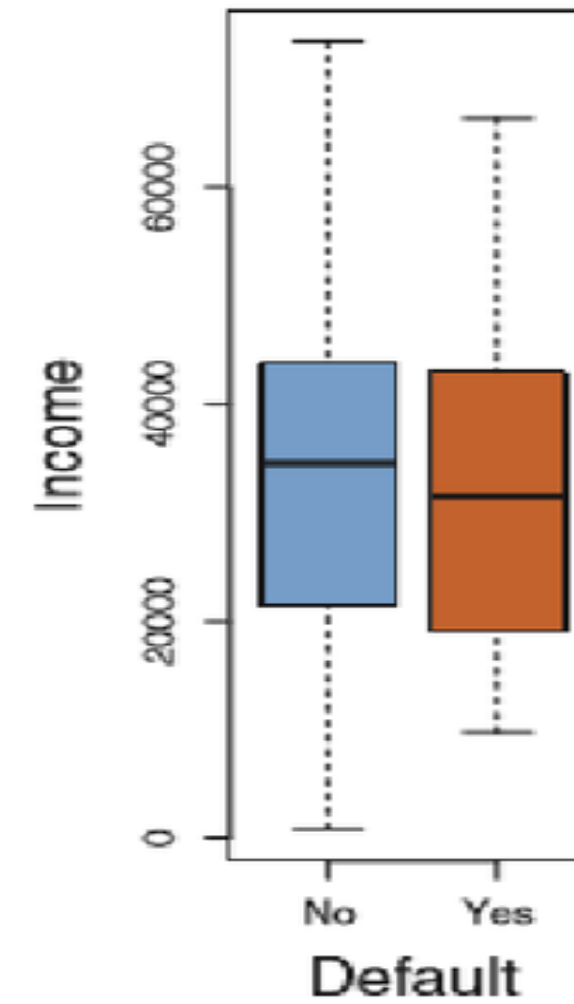
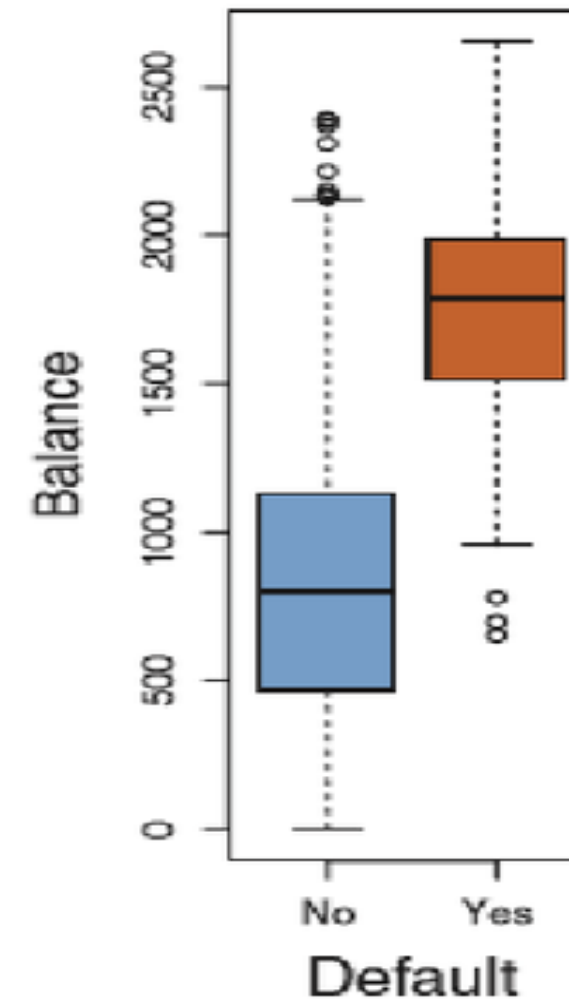
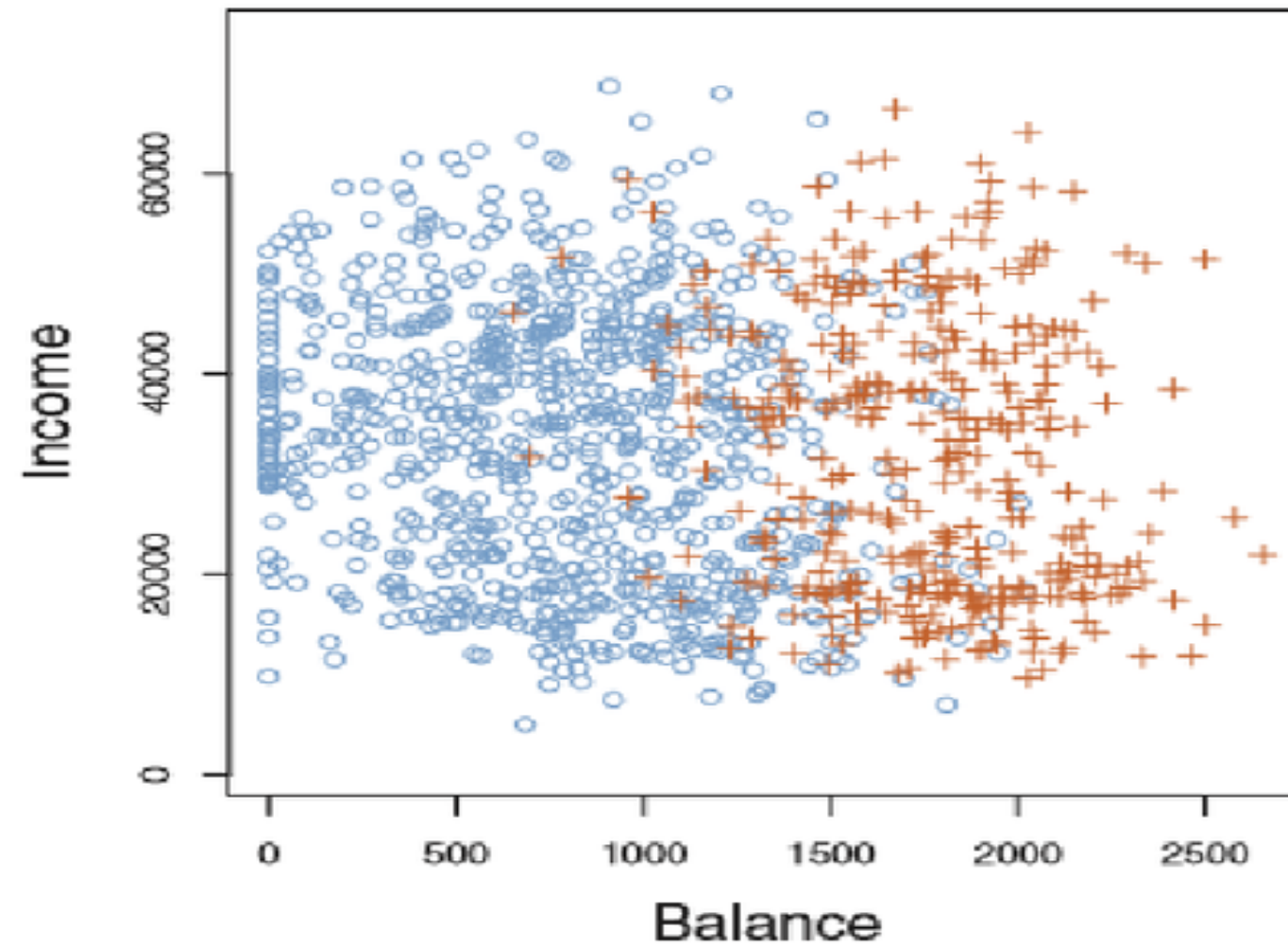
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- Which DNA mutations are deleterious and which are not?
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- We are still in the data scenario of $D = \{(\mathbf{x}_p, y_p)\}_{p=1}^P$ but $y_p \in \{0, 1\}$

Credit Default Prediction



- Easily Separable
- Find a function that determines class A vs. class B

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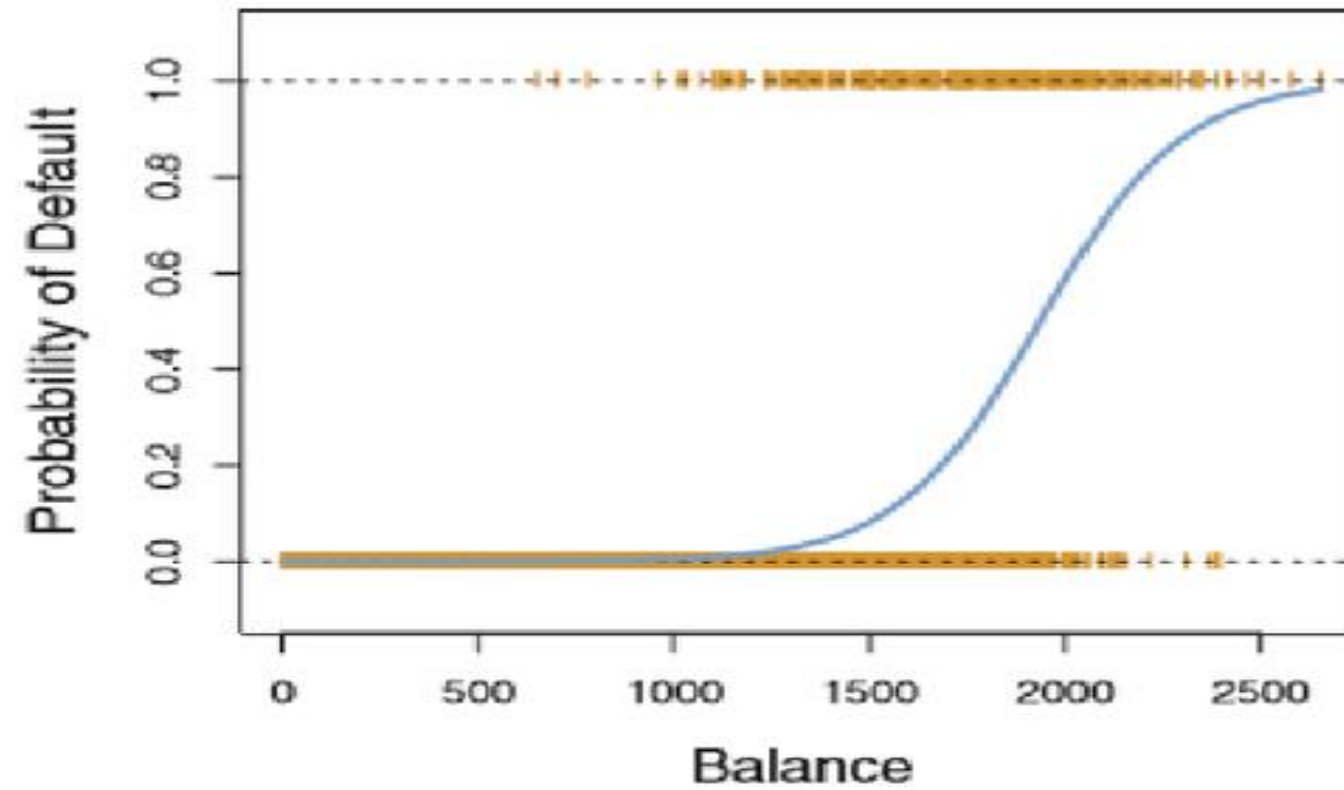
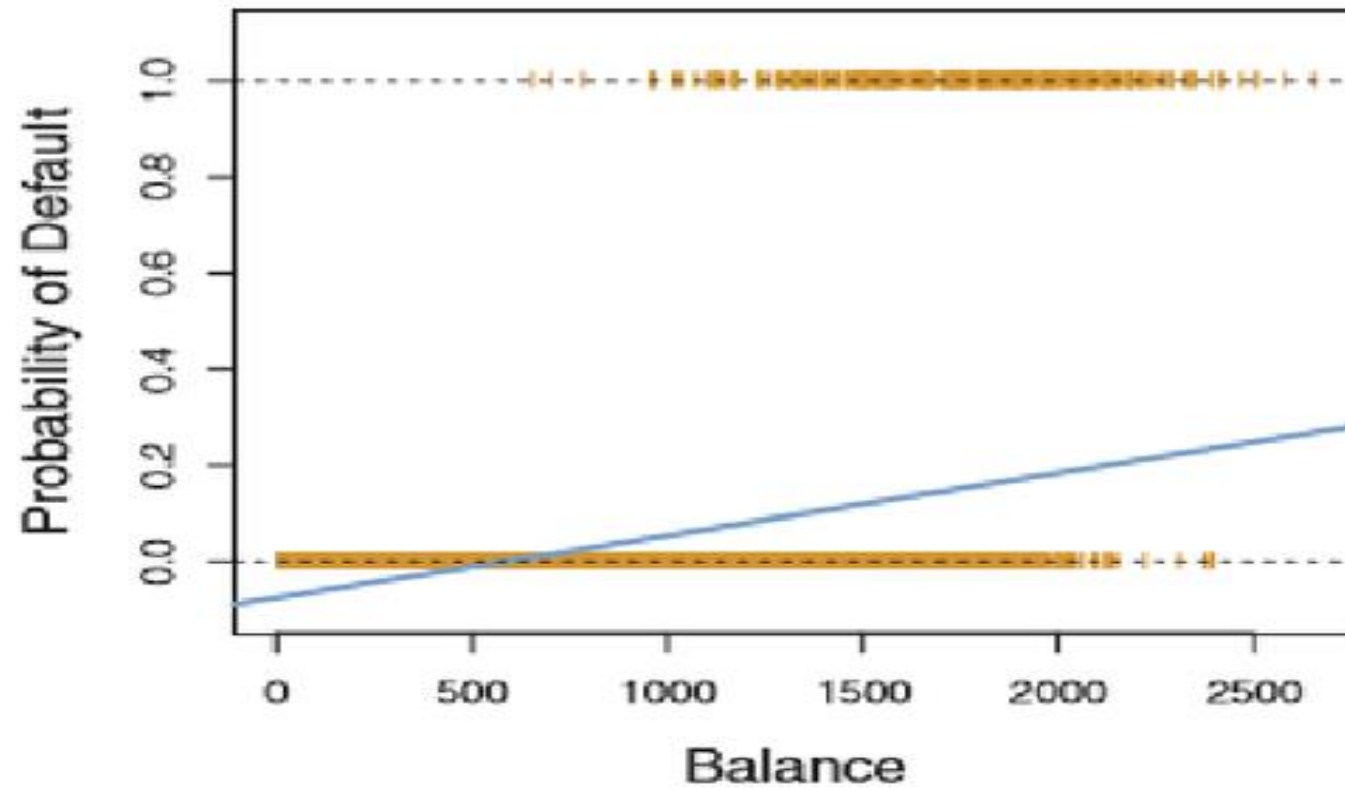
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- If a single binary decision, maybe linear regression works?

Linear vs. Logistic Regression



- Linear regression states $\hat{y} > 0.5$ as a decision rule - does that work?
- This is hard to interpret as $p(\text{default}|x)$
- What if we wanted to model probability of y belonging to a category instead of the actual response?

Takeaways

- What is Classification?
- Why not use Linear Regression?